

MedPath Diagnostics Pvt. Ltd.

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LABORATORY INVESTIGATION REPORT

Patient Name	Rajesh Kumar Sharma	Report No.	RPT-2025-04-00412
Age / Gender	44 Years / Male	Patient ID	MPD-DM-00412
Date of Test	08 April 2025	Date of Report	08 April 2025
Ref. Doctor	Dr. Anil Bhatia	Lab Technician	Mr. Pradeep Nair
Investigation	Type 2 Diabetes Screening Panel		

TYPE 2 DIABETES MELLITUS PANEL

Sr.	Test Name	Result	Normal Range	Units
1	Fasting Blood Glucose (FBG)	118 (H)	70 – 99	mg/dL
2	Post-Prandial Glucose (PPBG)	168 (H)	< 140	mg/dL
3	HbA1c (Glycosylated Haemoglobin)	6.2% (H)	< 5.7%	%
4	Fasting Insulin	14.2	2.6 – 24.9	µIU/mL
5	HOMA-IR (Insulin Resistance Index)	4.18 (H)	< 2.5	Index
6	Fasting C-Peptide	2.1	0.8 – 3.1	ng/mL
7	Urine Microalbumin	18	< 30	mg/24h
8	Serum Creatinine	0.92	0.6 – 1.2	mg/dL
9	eGFR (CKD-EPI)	87	≥ 60	mL/min/1.73m ²
10	Total Cholesterol	214 (H)	< 200	mg/dL
11	LDL Cholesterol	138 (H)	< 130	mg/dL
12	HDL Cholesterol	42	> 40 (Men)	mg/dL
13	Triglycerides	189 (H)	< 150	mg/dL

(H) = High (L) = Low Values outside normal range are flagged.

Remarks / Interpretation:

Fasting glucose of 118 mg/dL and HbA1c of 6.2% fall in the pre-diabetic range (impaired fasting glucose). Elevated HOMA-IR (4.18) indicates insulin resistance. Lipid panel shows mixed dyslipidaemia with elevated LDL, triglycerides, and total cholesterol. Recommend lifestyle modification and repeat HbA1c in 3 months. Physician review advised.

IRON DEFICIENCY ANAEMIA PANEL

Sr.	Test Name	Result	Normal Range	Units
1	Haemoglobin (Hb)	8.4 (L)	12.0 – 16.0 (F)	g/dL
2	RBC Count	3.42 (L)	3.8 – 5.2 (F)	×10 ⁶ /μL
3	Haematocrit (PCV)	28.6 (L)	36 – 48 (F)	%
4	MCV (Mean Corpuscular Volume)	71.2 (L)	80 – 100	fL
5	MCH	22.4 (L)	27 – 33	pg
6	MCHC	29.4 (L)	32 – 36	g/dL
7	RDW-CV	18.2 (H)	11.5 – 14.5	%
8	WBC Count	7,200	4,000 – 11,000	/μL
9	Platelet Count	2.8 × 10 ⁵	1.5 – 4.0 × 10 ⁵	/μL
10	Serum Iron	38 (L)	60 – 170	μg/dL
11	TIBC	468 (H)	250 – 370	μg/dL
12	Serum Ferritin	5.8 (L)	12 – 150 (F)	ng/mL
13	Transferrin Saturation	8.1% (L)	20 – 50	%
14	Reticulocyte Count	1.8	0.5 – 2.5	%
15	Serum Vitamin B12	342	200 – 900	pg/mL
16	Serum Folate	8.4	2.7 – 17.0	ng/mL

PBS: Microcytic hypochromic RBCs with anisopoikilocytosis noted.

Remarks / Interpretation:
Findings are consistent with moderate iron deficiency anaemia. Low haemoglobin (8.4 g/dL), markedly reduced ferritin (5.8 ng/mL), and low transferrin saturation (8.1%) with elevated TIBC confirm iron-depleted erythropoiesis. Vitamin B12 and folate are within normal limits. Likely cause: menorrhagia (as per clinical history). Oral iron supplementation with Vitamin C recommended. Repeat CBC and ferritin after 8 weeks.

Patient Name	Divya Raghunathan	Report No.	RPT-2025-04-00378
Age / Gender	38 Years / Female	Patient ID	MPD-THY-00378
Date of Test	13 April 2025	Date of Report	13 April 2025
Ref. Doctor	Dr. Sanjay Nambiar	Lab Technician	Ms. Deepa Narayanan
Investigation	Thyroid Function Tests (Pre-conception Screening)		

THYROID FUNCTION TEST (TFT) PANEL

Sr.	Test Name	Result	Normal Range	Units
1	TSH (Thyroid Stimulating Hormone)	6.8 (H)	0.5 – 4.0 (pre-conception)	mIU/L
2	Free T4 (fT4)	1.02	0.93 – 1.70	ng/dL
3	Free T3 (fT3)	3.1	2.3 – 4.2	pg/mL
4	Total T4	7.6	5.1 – 14.1	μg/dL
5	Anti-TPO Antibodies	128 (H)	< 35	IU/mL
6	Anti-Thyroglobulin Antibodies	48	< 115	IU/mL
7	TSH Receptor Antibodies (TRAb)	0.4	< 1.75	IU/L
8	Haemoglobin	12.8	12.0 – 16.0 (F)	g/dL
9	Serum Ferritin	44	12 – 150 (F)	ng/mL
10	Serum Vitamin D3 (25-OH)	22.4 (L)	30 – 100	ng/mL
11	Fasting Blood Glucose	88	70 – 99	mg/dL

Remarks / Interpretation:

TSH is mildly elevated at 6.8 mIU/L with normal fT3 and fT4, consistent with subclinical hypothyroidism. Elevated Anti-TPO antibodies (128 IU/mL) suggest autoimmune thyroid disease (Hashimoto's thyroiditis). Given pre-conception status, low-dose levothyroxine initiation is recommended to achieve TSH below 2.5 mIU/L before conception. Vitamin D3 supplementation advised. Repeat TFT in 4 weeks.

Lab Technologist

MedPath Diagnostics Pvt. Ltd.
Mumbai

Reporting Physician

Dr. Anita Krishnamurthy, DM (Endocrinology)
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